# Three-phase Circuits

# Homework

1. Solve for the line currents in the Y-Δ circuit of Fig.1. Take $Z_\Delta = 60\angle 45\,°\Omega$.

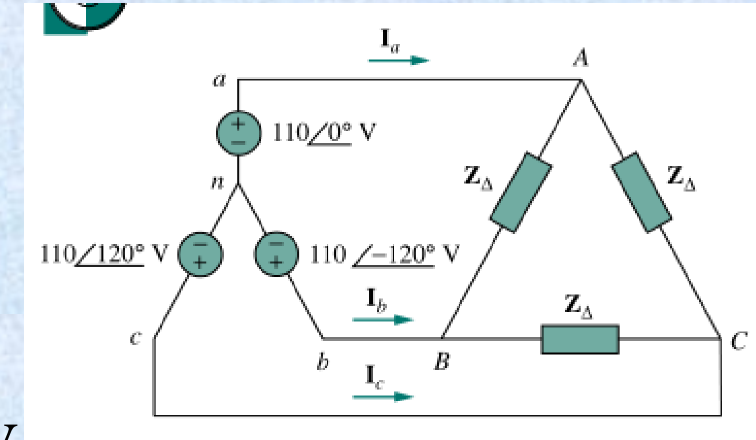
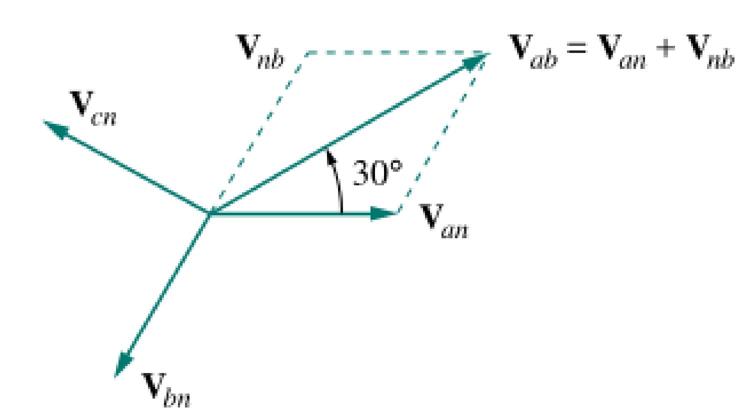



Figure 1

$$\dot{V}_{AB} = 110\angle 0° - 110\angle -120° = 110\sqrt{3}\angle 30°\text{V}$$

$$\dot{V}_{BC} = 110\angle -120° - 110\angle 120° = 110\sqrt{3}\angle -90°\text{V}$$

$$\dot{V}_{CA} = 110\angle 120° - 110\angle 0° = 110\sqrt{3}\angle 150°\text{V}$$




$$\dot{I}_{AB} = \frac{\dot{V}_{AB}}{Z_\Delta} = \frac{110\sqrt{3}\angle 30°}{60\angle 45°} \approx 3.175\angle -15°\text{A}$$

$$\dot{I}_{BC} = \frac{\dot{V}_{BC}}{Z_\Delta} \approx 3.175\angle -135°\text{A}$$

$$\dot{I}_{CA} = \frac{\dot{V}_{CA}}{Z_\Delta} \approx 3.175\angle 105°\text{A}$$

$$\dot{I}_a = \dot{I}_{AB} - \dot{I}_{CA} = 3.175(\angle -15° - \angle 105°)$$

$$= 3.175\sqrt{3}\angle -45° = 5.5\angle -45°\text{A}$$

$$\dot{I}_b = 5.5\angle -165°\text{A} \qquad \dot{I}_c = 5.5\angle 75°\text{A}$$

$Z_\Delta = 60\angle 45\,°\Omega$

$Z_Y = Z_\Delta / 3 = 20\angle 45\,°\Omega$

$$\dot{I}_a = \frac{110\angle 0°}{20\angle 45°} = 5.5\angle -45°\text{A}$$

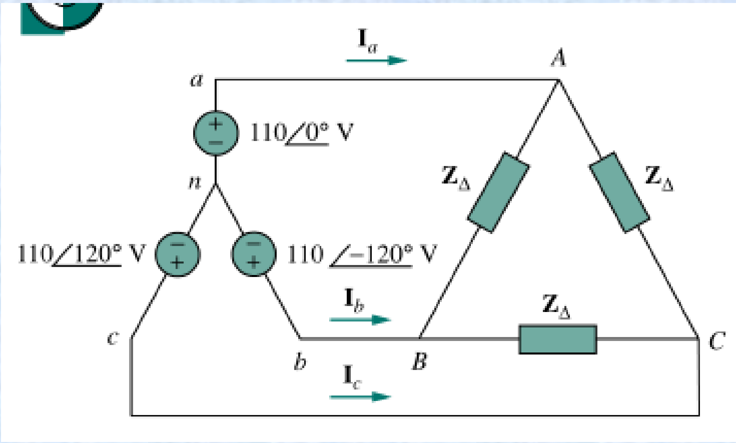
(3.89–j3.89)

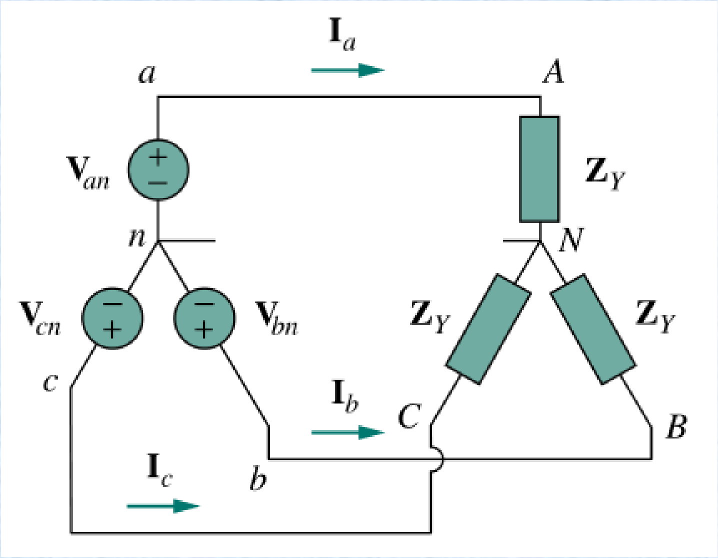
$$\dot{I}_b = \frac{110\angle -120°}{20\angle 45°} = 5.5\angle -165°\text{A}$$

(–5.31–j1.42)

$$\dot{I}_c = \frac{110\angle 120°}{20\angle 45°} = 5.5\angle 75°\text{A}$$

(1.42+j5.31)



2. Obtain the line currents in the three-phase circuit of Fig.2.

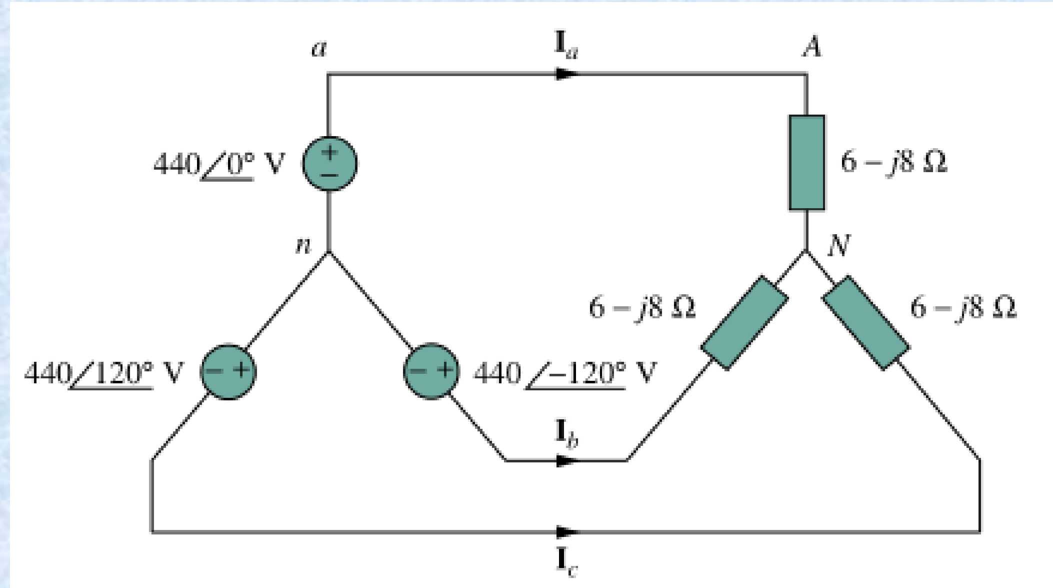



Figure 2

$$Z = 6 - j8 = 10\angle -53.13°\Omega$$

$$\dot{I}_a = \frac{440\angle 0°}{10\angle -53.13°} = 44\angle 53.13°\text{A}$$

(26.4+j35.2)

$$\dot{I}_b = \frac{440\angle -120°}{10\angle -53.13°} = 44\angle -66.87°\text{A}$$

(17.28–j40.46)

$$\dot{I}_c = \frac{440\angle 120°}{10\angle -53.13°} = 44\angle 173.13°\text{A}$$

(–43.68+j5.26)

3. For the three-phase circuit in Fig. 3, find the average power absorbed by the delta-connected load with $Z_\Delta = 21 + j24\Omega$.

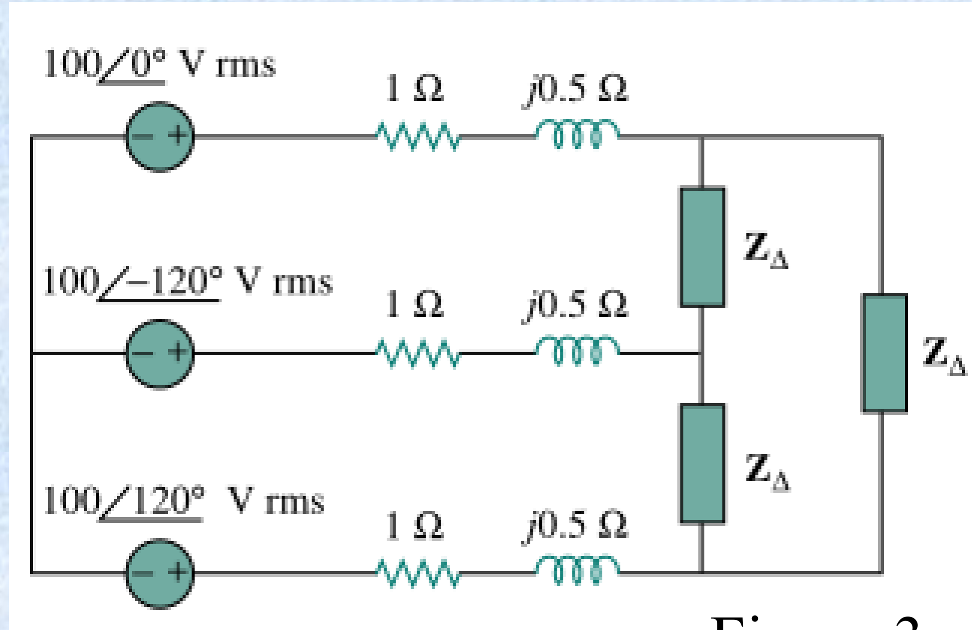



Figure 3

$Z_Y = Z_\Delta/3 + 1 + j0.5 = 8 + j8.5\ \Omega$

$= 11.67\angle 46.74°\ \Omega$

$$\dot{I}_a = \frac{100\angle 0°}{11.67\angle 46.74°} \approx 8.57\angle -46.74° \mathrm{A}$$

$$P_a = 7I_a^{\,2} = 7 \times 8.57^2 = 513.76\mathrm{W}$$

$$P = 3P_a = 1541.28\mathrm{W}$$